

Timeline and Conditions tested:

1. Set up per task:

Open Field:

- Task set up:
 - open-field box (50cmx50cm) without lid
- Camera set up: mounted at the top center, 5 cm above the box
 - Camera and lens:
 - DMK22AUC03, 744×480 (0.4 MP) Y800 (*Camera 1)
 - Computar TG4Z2813FCS-IR, 1/3-inch 2.8-12mm f1.3 Auto Iris, Day/Night IR (*Lens 1)
 - Setting on IC Capture:
 - Holder position: Top-down mounting the camera

Pole Task:

- Task set up:
 - open-field box (50cmx50cm) without lid
 - a vertical rough-surfaced pole (50 cm height, 0.7 cm diameter) fixed on a stable base.
- Camera set up: mounted at the top center, 5 cm above the box
 - Camera and lens:
 - Camera 1
 - Lens 1
 - Setting on IC Capture:
 - Holder position: Top-down mounting the camera

Grid task:

- Task set up:
 - a metal frame with a horizontal wire mesh grid (20.32 cm x 60.96 cm, 1.27cm x 1.27cm rung, 0.1cm diameter wire) secured at its mid-height
 - transparent boards enclosing the upper part of the metal frame
 - mirror put in the lower part of the frame
- Camera set up:
 - Camera and lens:
 - DMK22AUC03, 744×480 (0.4 MP) Y800 (*Camera 2)
 - Computar TG3Z2910AFCS-IR, 1/3-inch 2.9-8.2mm f1.3 Auto Iris, Day/Night IR (*Lens 2)
 - Setting on IC Capture:

- Holder position: mounting in front of and facing the frame

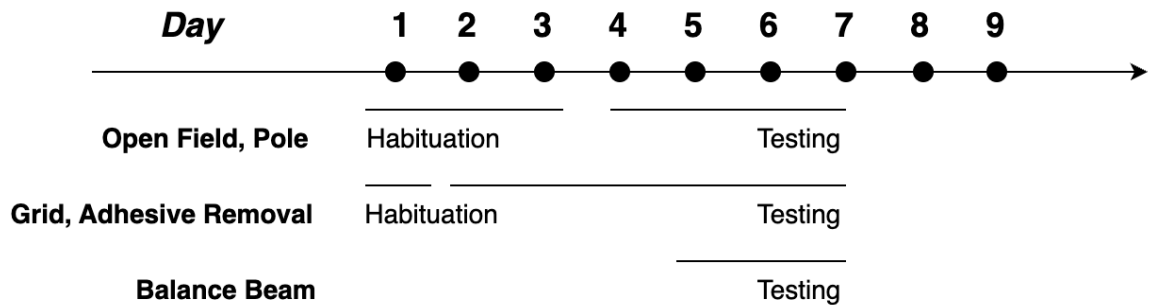
Adhesive removal task:

- Task set up:
 - 15cmx15cmx15cm clear acrylic box
 - a mirror box positioned behind the box, reflecting a view of its inside
 - black and white adhesive medical tapes (0.3cmx0.4cm)
- Camera set up:
 - Camera and lens:
 - Camera 2
 - Lens 2
 - Setting on IC Capture:
 - Holder position: mounting in front of and facing the box

Balance beam 25mm and 5mm:

- Task set up:
 - 25mm beam
 - 5mm beam
 - a box was placed at one end of the beam to serve as the mouse's destination
- Camera set up:
 - Camera and lens:
 - Camera 1 & 2
 - Lens 1 & 2
 - Setting on IC Capture:
 - Holder position: Camera 1 with Lens 1 was mounted at the side of the beam to capture its full length. Camera 2 with Lens 2 was mounted at the beam's starting end to record from the front view

2. Overall timeline per cohort:



Task protocol:

The tasks should be done in the order of open field, pole task, grid task, and finally adhesive removal task, and keep the same order for all the mice if you're doing multiple mice a day.

Open Field:

- *Timeline:* The task requires 3 days of habituation and 3+ days of testing. The first 3 days of the task are for habituation. The testing period starts on the 4th day.
- *Total time/mouse/day:* 20 minutes.
- *Task protocol:* The mouse is allowed to explore the open field (50cmx50cm) for 20 minutes.
- *Analysis:* The body position is tracked using MATLAB to convert the pixel change in each frame of the video, and spot out the largest and darkest moving object in the frame. The body center point is then converted into 2D coordinates (x,y) in order to calculate basic locomotion kinetics such as speed, acceleration, time spent stopping and running, and angular velocity. Instantaneous speed per day is plotted in histogram and per minute in order to evaluate the tracking code.

Pole Task:

- *Timeline:* The task requires 3 days of habituation and 3+ days of testing. The first 3 days of the task are for habituation. The testing period starts on the 4th day.

- *Total time/mouse/day*: maximum 18 minutes total time, maximum 4 minutes per trial, 4 trials in total.
- *Task protocol*: A pole of roughly 1 cm ($\frac{3}{8}$ inches) in diameter is placed in the open field. The mouse is then placed facing up on the pole and allowed to turn and descend, roam around in the open field for 15 seconds, and the process is repeated for a total of 4 times. A cool white light (6500K) LED lamp of 4 watts is placed shining directly on top of the pole to encourage descent. The maximum duration of 120s each for the downward turn period and descend period (downward orientation until all four paws reached the base of the pole).
 - The open field and the pole task share a camera. Do not move the camera as it is set up for both tasks. If you happen to move it, there's a piece of tape at each knob of the camera holder to secure it, just move the hand of the camera holder back to where the tapes and the lines on the tapes are straight.
- *Analysis*: The following are measured: the time for the mouse to turn from an upright position/orientation to a downward position and the time for the mouse to descend the pole from said downward position. The maximum duration of 120 s each for the downward turn period and descend period (downward orientation until all four paws reached the base of the pole). If the mouse is unable to turn or descend, or fall from the pole, the time recorded for that trial would be 120 for each period.

Grid Task:

- *Timeline*: The task requires 1 day of habituation and 3+ days of testing.
- *Total time/mouse/day*: 5 minutes.
- *Task protocol*: The mouse is placed on a metal grid (20.32 cm x 60.96 cm, 1.27cm x 1.27cm rung, 0.1cm diameter wire) and allowed to walk around for 3 minutes. There is a mirror piece secured at a 45 degree angle below the grid to see the foot slips.
 - Don't move the mirror position. It is also secured by double sided tapes that are pretty hard to move.
- *Analysis*: 50 steps are randomly picked out from the video by randomly picking out 50 0.5 seconds non overlapping video of when the mouse is locomoting (above 0.5cm/s). The number of complete forelimb foot slips are manually calculated by examining each video until the number of

steps counted reach 50 steps. The number of foot slips are normalized by the mouse's correct steps on the grid.

Adhesive Removal Task:

- *Timeline:* The task requires 1 day of habituation and 3+ days of testing.
- *Total time/mouse/day:* 30 minutes first day, maximum 6 minutes following days.
- *Task protocol:* The set up includes a 15cmx15cmx15cm clear acrylic box with a mirror piece leaned against it, behind it. On the first day of the task, the mouse is placed in the box for 30 minutes for habituation. On the following testing days, the mouse is placed in the clear box for 1 minute. Then, it is scruffed and 2 strips of different color adhesive medical tapes (0.3cmx0.4cm) are placed with equal pressure on the hairless part of the forepaws (i.e., the three pads, thenar and hypothenar). The mouse is placed back into the beaker. The total time to remove the adhesive tape for both paws is measured when the mouse feels the tape (mouse bringing its paws to its mouth or rubbing its two palms together).
 - The grid task and the adhesive removal task share a camera. You have to move the camera position between the 2 tasks. Adjust only the first knob of the camera holder, do not adjust the second knob (that is attached to the camera), nor change the focus on the camera. You can adjust the first knob by turning the lock to loosen it, and move the hand of the camera holder only up or down, don't twist it.
 - Grid task: Move the acrylic box and mirror from the adhesive task out of the way. There are 2 blue tape pieces with black marker lines on the camera holder for the angle of the hand of the camera holder for the grid task. Once the adhesive task is done, adjust the first knob so that the lines on the 2 pieces of tape make a straight line (you can test this with a ruler). That is the camera position for the grid task. There is also a long piece of white tape on the table to help you align the bottom of the frame to it as additional help for the camera position.
 - Adhesive task: There are 2 pieces of tape (white and blue) forming a 90 degree angle on the table. Move the corner of the acrylic box to that angle, and one of the sides of the box to align the white tape piece. There are 2 blue pieces of tape on the mirror, align those 2

pieces with the sides of the acrylic box, lean the mirror against it behind it. All tape pieces provide resistance to the moving pieces (acrylic box and mirror) so it is pretty stable. If you happen to knock anything out of place, just realign them. Once the grid task is done, adjust the first knob down to the position for the adhesive task so that the bottom of the video frame aligns with where the white tape and sides of the box touch.

- *Analysis:* The following are measured: the time for the mouse to recognize the tape (mouse bringing its paws to its mouth or rubbing the two tapes together), the time for the mouse to remove the first tape, and the time for the mouse to remove the second tape. Maximum duration given to the mouse to remove the two adhesives was 300s. You can usually spot the tape pieces on the surface of the box once they are removed, if not, they are on the mouse body.

Balance beam 25mm + 5mm:

- *Timeline:* The task is done on the 7th-9th days of the timeline or at the end of the habituation period for all the other tasks.
- *Total time/mouse/day:* 5min (or less)
- *Task protocol:* put the mouse on the 25mm beam and let it cross over to the other side (with a dark box + with the bedding from that cage to encourage crossing) 5 times. This usually happens pretty fast and since the mouse can be a bit freaked out, be on alert to catch it (there is a mesh thing underneath the beam so you can catch the mouse). Afterwards, switch out the 25mm beam for the 5mm beam. Put the mouse back on the beam and let it cross once (twice if it didn't cross the first time- because of falling or just didn't move- if it didn't move at all, just note down). The beam has a rough surface surrounding (velcro) it so that the mouse can walk on it still. Mice can learn to walk on the beam relatively fast so make sure to not overtrain since you might not be able to see the deficit you expect to see.
 - The beam has 2 cameras. The front view camera is shared with the grid and sticker task. You have to turn the camera 180 because the beam is on the opposite side of those two tasks. The side view camera is positioned on the left side of the table, at the starting point of the beam task.
- *Analysis:* The front view camera is for calculating crossing, pausing and potentially crawling time as the mouse crosses the beam. The side view camera is for calculating the number of slips on the beam (by eye).